

Module 10: Fractal Geometry and the Limits of Prediction — Summary

What this module is about. A recursive rule can be completely deterministic and still produce very different kinds of long-term behavior. Some rules keep nearby starting values close. Some amplify differences in a predictable way. Chaotic rules can make a tiny uncertainty in the starting value grow into a large difference later. Fractals add another idea: measurements can depend strongly on the scale at which a boundary is observed.

A recursion has the form

$$x_{n+1} = f(x_n).$$

The same rule is applied repeatedly. The logistic map is

$$x_{n+1} = rx_n(1 - x_n).$$

For some values of r , the sequence settles toward a fixed value; for other values, it becomes periodic or chaotic. In a chaotic regime, nearby starting values can eventually produce very different forecasts.

Example 1: Collatz trajectories

The Collatz rule is

$$n \mapsto \begin{cases} n/2, & n \text{ even,} \\ 3n + 1, & n \text{ odd.} \end{cases}$$

Starting from 7:

$$7 \rightarrow 22 \rightarrow 11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

This path takes 16 steps and reaches a peak of 52. Nearby starting integers can have very different trajectories even though the rule itself is simple and exact.

For the Koch curve, each segment is replaced by 4 segments of one-third the original length. Therefore the total length is multiplied by

$$4 \left(\frac{1}{3} \right) = \frac{4}{3}$$

at each step. Starting from length L_0 ,

$$L_n = L_0 \left(\frac{4}{3} \right)^n.$$

For the Sierpinski triangle, $3/4$ of the previous area remains at each step:

$$A_n = A_0 \left(\frac{3}{4} \right)^n.$$

Thus boundary complexity can increase while remaining area decreases.

Example 2: Scale-dependent coastline measurement

Suppose a coastline is measured with rulers of different lengths.

ruler length	placements	measured length
10	50	500
5	130	650
2.5	340	850
1.25	900	1125

Shorter rulers follow smaller bends in the boundary and produce a larger measured length. The coastline has not physically grown; the observation scale has changed.

Example 3: A first logistic-map step

With $r = 2.8$ and $x_0 = 0.2$,

$$x_1 = 2.8(0.2)(1 - 0.2) = 2.8(0.2)(0.8) = 0.448.$$

The rule is deterministic: once r and x_0 are fixed, the next value is fixed. But determinism alone does not guarantee that long-range forecasts remain stable when the initial measurement is slightly uncertain.

By the end of this module, you should be able to: iterate simple recursive rules; compare Collatz trajectories; compute Koch segment counts and total-length factors; compute Sierpinski area fractions; carry out logistic-map steps; and explain with a concrete calculation why some measurements and predictions depend on scale or initial conditions.

Module 10: Fractal Geometry and the Limits of Prediction — Prequiz

Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.

Name:

Date:

Useful rules

Collatz: even $n \mapsto n/2$, odd $n \mapsto 3n + 1$. Koch length factor: $4/3$. Sierpinski area factor: $3/4$. Logistic map: $x_{n+1} = rx_n(1 - x_n)$.

1. Apply the Collatz rule starting from 9 until the sequence reaches 1.
 - (a) Write the full sequence.
 - (b) How many steps are required to reach 1?
 - (c) What is the largest value reached?

2. A Koch curve begins with one segment of length 1.
 - (a) How many segments are present at step 3?
 - (b) What is the total length at step 3? Give an exact fraction and a decimal approximation.
 - (c) By what factor does total length change from one step to the next?

3. A Sierpinski triangle keeps $3/4$ of its area at each step. What fraction of the original area remains after 4 steps? Give both the exact expression and a decimal approximation.

4. Use the logistic map

$$x_{n+1} = 2.5x_n(1 - x_n)$$
 with $x_0 = 0.3$. Compute x_1 . Then explain what information would be needed to determine x_2 .

5. A surveyor measures a coastline with the data below.

ruler length (km)	number of placements
8	60
4	135
2	300
1	680

For each row, compute ruler length $\times$ placements. Describe the numerical trend as the ruler gets shorter. Then explain why the trend comes from the measurement scale rather than from a physical change in the coastline.

Module 10: Fractal Geometry and the Limits of Prediction — Matching Quiz

These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.

Name:

Date:

1. Apply the Collatz rule starting from 6 until reaching 1. Record the sequence, number of steps, and peak value.
2. A Koch curve starts with one segment of length 1. How many segments are present at step 4, and by what factor has total length changed from step 0 to step 4?
3. What fraction of the original Sierpinski-triangle area remains after 3 steps? Give a decimal approximation.
4. For $x_{n+1} = 3x_n(1 - x_n)$ with $x_0 = 0.5$, compute x_1 and x_2 .
5. A coastline is measured with increasingly short rulers, and the computed length increases each time the ruler gets shorter. What mathematical idea from this module does that demonstrate? Explain in one or two clear sentences.